

## Before global or local rotations

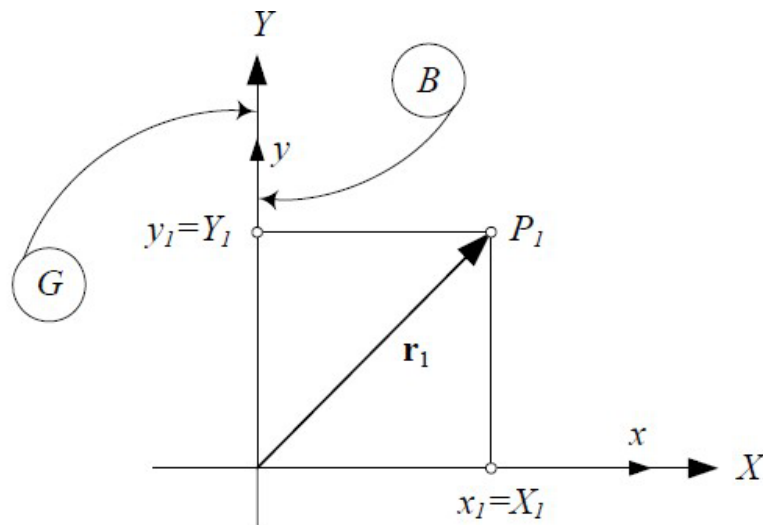


FIGURE 2.2. Position vector of point P when local and global frames are coincident.

## Rotation about global axis

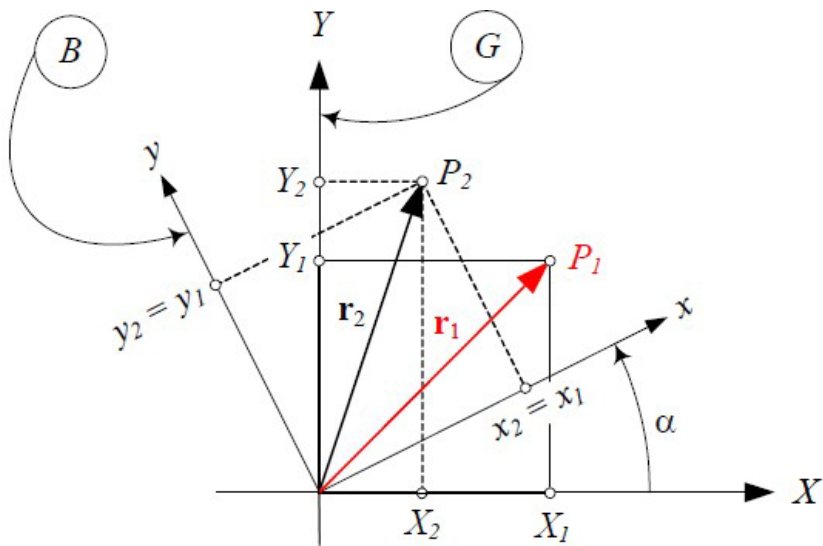


FIGURE 2.4. Position vectors of point P before and after the rotation of the local frame about the Z-axis of the global frame.

$${}^G \mathbf{r} = Q_{Z,\alpha} {}^B \mathbf{r} \quad (2.1)$$

## Rotation about local axis

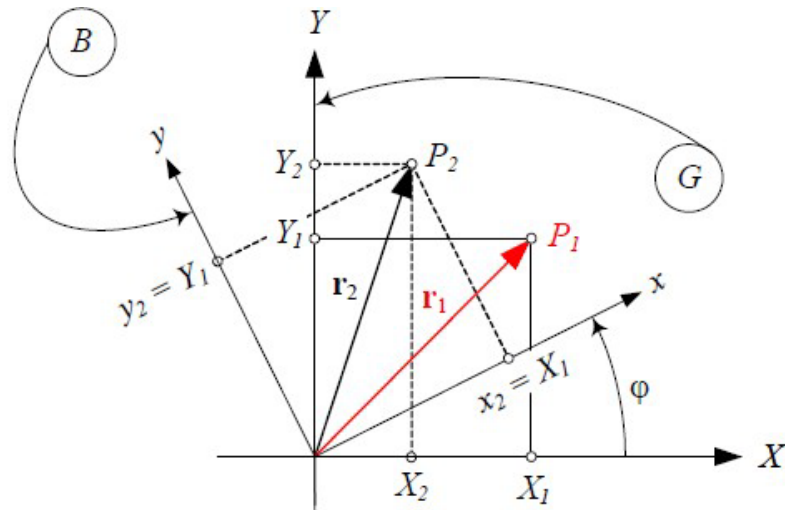


FIGURE 2.10. Position vectors of point  $P$  before and after rotation of the local frame about the  $z$ -axis of the local frame.

$${}^B\mathbf{r} = A_{z,\varphi} {}^G\mathbf{r}. \quad (2.69)$$

where  ${}^B\mathbf{r}$  is local coordinates (local coordinates remain the same before and after rotation),  ${}^G\mathbf{r}$  is after-rotation global coordinates.  
( ${}^G\mathbf{r}$  before rotation is the same as  ${}^B\mathbf{r}$ )

With only one rotation, equ (2.1) and equ (2.69) are equivalent, there is no difference between one global and one local rotation. When there are successive rotations, the difference shows: in global rotations, the global frame is used as current (reference) frame, while in local rotations the local frame is used as current (reference) frame.

## Comparisons of Successive Global Rotations and Success Local Rotations

|          | Rotation (local frame) about global axis                                                                                                                                                                                             | Rotation (local frame) about local axis                                                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Problem  | Given local coordinate, find the global coordinate after rotating the local frame                                                                                                                                                    | Given the global coordinate after rotation, find its local coordinates                                                                                             |
| Equation | ${}^G\mathbf{r} = Q * {}^B\mathbf{r}$ <p>where <math>Q</math> is the rotation matrix</p> <p>Multiply <math>Q^{-1}</math> to both sides, we get:</p> ${}^B\mathbf{r} = Q^{-1} * {}^G\mathbf{r}$ ${}^B\mathbf{r} = A * {}^G\mathbf{r}$ | ${}^B\mathbf{r} = A * {}^G\mathbf{r}$ <p>where <math>A = Q^{-1}</math></p> <p><math>{}^G\mathbf{r}</math> is the global coordinate after the local rotation</p>    |
|          | After rotation of the local frame about the global axis, we are able to find the global coordinates of a point of a rigid body, if we have its local                                                                                 | After rotation of the local frame about the local-axis, we are able to find the coordinates of any point of a rigid body in local coordinate frame, if we have its |

|  |                        |                                             |
|--|------------------------|---------------------------------------------|
|  | Coordinates (Fig 2.4). | coordinates in the global frame (Fig 2.10). |
|--|------------------------|---------------------------------------------|

With one rotation only, there is no difference between rotations about global axis and local axis.

**It makes a difference when there are successive rotations:**

- Rotation of the local frame about global axis is to use **global frame** as reference.
- Rotation of the local frame about local axis is to use **local frame (current frame)** as reference

## Successive Rotations

| Successive ( <i>local frame</i> ) Rotations about Global Axis                                                                                                                                                                                                                                                                                                                 | Successive ( <i>local frame</i> ) Rotations about Local Axis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initial:<br>Global frame and local frame are <b>coincident</b> : ${}^B r = {}^G r$<br><br>For i=1 to n<br>${}^G r = Q_i * {}^B r$<br><b>Reset local frame be coincident with global frame</b><br>Update ${}^B r \leftarrow {}^G r$<br>endfor<br><br>Result: (Rotation about global frame)<br>${}^G r = Q_n * Q_{n-1} * \dots * Q_2 * Q_1 {}^B r$<br><b>Pre-multiplication</b> | Initial:<br>Global frame and local frame are <b>coincident</b> : ${}^B r = {}^G r$<br><br>For i=1 to n<br>${}^B r = A_i * {}^G r$<br><b>Set local frame as reference frame</b><br>Update ${}^G r \leftarrow {}^B r$<br>endfor<br><br>Result: (Rotations about current frame (local frame))<br>${}^B r = A_n * A_{n-1} * \dots * A_2 * A_1 * {}^G r$<br>Multiple $(A_n * A_{n-1} * \dots * A_2 * A_1)^{-1}$ to both sides:<br>${}^B r = (A_n * A_{n-1} * \dots * A_2 * A_1)^{-1} {}^G r$<br><b>Global coordinates after successive local rotations:</b><br>${}^G r = A_1^{-1} * A_2^{-1} * \dots * A_{n-1}^{-1} * A_n^{-1} * {}^B r$<br><br>${}^G r = Q_1 * Q_2 * \dots * Q_{n-1} * Q_n * {}^B r$<br><b>Post-multiplication</b> |

Example: Given a position  $r_p = [1 \ 1 \ 1]^T$ , find its global coordinate after the following rotations in sequence.

- (1) Rotate about Z-axis 30 degrees
- (2) Rotate about z-axis 60 degrees
- (3) Rotate about x-axis 45 degrees
- (4) Rotate about Y-axis 30 degrees

Solution:

Initially local and global frames are coincident.  ${}^B r_p = [1 \ 1 \ 1]^T$

Without any rotations,  ${}^G r_p = I * {}^B r_p$  where I is an identity matrix.

Q1(Z, 30d), Q2(z, 60d), Q3(x, 45d), Q4(Y, 30d)

The subscript after Q indicates the order of the rotations.

After the first rotation (about Global axis):

pre-multiplication  $Q = Q1(Z, 30d) * I$

After the 2<sup>nd</sup> rotation (about Local axis):

post-multiplication  $Q = Q1(Z, 30d) * I * Q2(z, 60d)$

After the 3<sup>rd</sup> rotation (about Local axis):

$$\text{post-multiplication } Q = Q1(Z, 30d) * I * Q2(z, 60d) * Q3(x, 45d)$$

After the 4<sup>th</sup> rotation (about Global axis): pre-multiplication

$$Q = Q4(Y, 30d) * Q1(Z, 30d) * I * Q2(z, 60d) * Q3(x, 45d)$$

$$Q1(Z, 30d) = \begin{bmatrix} c30d & -s30d & 0 \\ s30d & c30d & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad Q2(z, 60d) = \begin{bmatrix} c60d & -s60d & 0 \\ s60d & c60d & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Q3(x, 45d) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c45d & -s45d \\ 0 & s45d & c45d \end{bmatrix}$$

$$Q4(Y, 30d) = \begin{bmatrix} c30d & 0 & s30d \\ 0 & 1 & 0 \\ -s30d & 0 & c30d \end{bmatrix}$$

Plug in above rotation matrices and  ${}^B r_p = [1, 1, 1]^T$  into the equation below

$${}^G r_p = Q * {}^B r_p = Q4(Y, 30d) * Q1(Z, 30d) * I * Q2(z, 60d) * Q3(x, 45d)$$

Solving with Octave:

```
octave:12> r_B=[1, 1,1]'
```

```
r_B =
```

```
1  
1  
1
```

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```
>>> Q1_Z=[cosd(30), -sin(30), 0; sind(30), cosd(30), 0; 0, 0,1]
```

```
octave:6> Q1_Z=[cosd(30), -sind(30), 0; sind(30), cosd(30), 0; 0, 0,1]
Q1_Z =
```

```
    0.8660   -0.5000         0
    0.5000    0.8660         0
         0         0    1.0000
```

```
octave:7> Q2_z=[cosd(60), -sind(60), 0; sind(60), cosd(60), 0; 0, 0,1]
Q2_z =
```

```
    0.5000   -0.8660         0
    0.8660    0.5000         0
         0         0    1.0000
```

```
octave:8> Q3_x=[1, 0,0; 0, cosd(45), -sind(45); 0, sind(45), cosd(45)];
```

```
octave:9> Q3_x=[1, 0,0; 0, cosd(45), -sind(45); 0, sind(45), cosd(45)]
Q3_x =
```

```
    1.0000         0         0
         0    0.7071   -0.7071
         0    0.7071    0.7071
```

```
octave:10> Q4_Y=[cosd(30), 0, sind(30); 0, 1, 0; -sind(30), 0, cosd(30)]
Q4_Y =
```

```
    0.8660         0    0.5000
         0    1.0000         0
   -0.5000         0    0.8660
```

```
octave:15> Q= Q4_Y*Q1_Z*Q2_z*Q3_x
Q =
```

```
    4.3453e-17   -2.5882e-01    9.6593e-01
    1.0000e+00    4.4511e-17   -4.4511e-17
   -2.0318e-17    9.6593e-01    2.5882e-01
```

```
octave:16> r_G=Q*r_B
r_G =
```

```
    0.7071
    1.0000
    1.2247
```